Locality Conditions on *Wh*-Movement in the Miracle Creed Framework

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1. Introduction

Since Ross's (1967) discovery of island conditions, unification of these has been a significant issue in generative syntax. Chomsky's (1973) Subjacency Condition is the first important step toward unifying Ross's (1967) islands such as the *Wh*-Island Condition and the Complex NP Condition into a single constraint attributed to UG. Huang's (1982) Condition on Extraction domain (CED) tried to unify the Subject Condition and the Adjunct Condition into one, and Chomsky's (1986) *Barriers* attempted to unify the Subjacency Condition and the CED into the notion of Barrier. What is common to these approaches is the idea that a certain domain constitutes an absolute bounding domain for movement. In contrast, Rizzi's (1990) Relativized Minimality, which attempted to unify several locality conditions including the *Wh*-Island Condition, states that movement cannot be applied across a 'closer' intervening position that belongs to the same type with the landing site, with closeness defined in terms of asymmetric c-command.[1] In Minimalism, the idea that a certain category serves as an absolute barrier for extraction is handed over to Phase Theory (Chomsky 2000, 2001, 2008) and the idea of intervention in terms of c-command to the Minimal Link Condition (Chomsky 1995) and Probe-Goal Search (Chomsky 2000, 2001), though it remains to be seen whether island phenomena noted above can be unified and whether two types of notion of locality conditions, domain-based and intervention-based approaches, coexist or one replaces another.[2]

This article aims to unify island conditions on *wh*-movement in English based on the latest minimalist framework, the Miracle Creed (MC) framework proposed by Chomsky (2024) and extended by Kitahara and Seely (2025), according to which A'-movement is reinterpreted as phase-level access from C at the Conceptual-Intentional (CI) interface and externalization. This article argues that the Subject Condition, the Adjunct Condition, the *Wh*-Island Condition, and the Complex NP Condition in English is unified under two locality conditions on phase-level access, the Box-over-Box Condition and the Minimality Condition on Access. Informally speaking, the former states that once search initiated by C has located a 'boxed' item (i.e., an item accessible to C), then it never 'look inside' the 'box.' By contrast, the latter prohibits C from 'looking inside' the search domain of another C.[3]

This article is organized as follows. In section 2, we will review the MC framework by Chomsky

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[1] This idea goes back to Chomsky's (1973) Superiority Condition.

[2] See Dikken and Lahne (2013) and Hu (2024, Chap. 2) for historical development of syntactic theories on locality. For a survey of unified and non-unified approaches to island phenomena, see Boeckx (2012)

[3] This article pays attention only to English, leaving cross-linguistic variations in locality conditions for future inquires.

(2024) and Kitahara and Seely (2025). Based on this framework, we will propose in section 3 two locality conditions on phase-level access, the Box-over-Box Condition and the Minimality Condition on Access, demonstrating that these two conditions provide a principled explanation for locality conditions on *wh*-movement in English such as the Subject Condition, the Adjunct Condition, the *Wh*-Island Condition, and the Complex NP Condition. Section 4 will extend the proposed analysis to the superiority effect, arguing that multiple *wh*-questions in English is not derived by phase-level access from C_Q but by access from a *wh*-phrase already accessed by C_Q. Section5 is a conclusion.

2. The MC Framework

This section reviews the MC framework. In section 2.1. we introduce Chomsky's (2024) proposal that A'-movement is replaced by phase-level access at CI and externalization, with the leading ideas behind it briefly summarized. In section 2.2, we outline the framework revised by Kitahara and Seely (2025), which overcomes some empirical and theoretical problems with Chomsky's (2025) original framework.

2.1 Chomsky (2024)

Chomsky (2024) proposes Principle [S] and [T] in (1) as a guideline for inquiry.

(1) a. Principle [S]: The computational structure of language should adhere as closely as possible to SMT. (Chomsky (2024:19))

b. Principle [T]: All relations and structure-building operations (SBO) are thought-related, with semantic properties interpreted at CI. (Chomsky (2024:22))

Principle [S] states that the faculty of language conforms the Strong Minimalist Thesis (SMT), which requires that structures of I-language are generated by maximally simple operations governed by third-factor conditions understood in the context of natural law (Chomsky 2005). Principle [T] is the idea that language is a thought-generating system, and every structure-building operation is associated with interpretation at CI. Guided by Principle [S] and [T], third-factor principles and conditions specific to human language (Language Specific Conditions, LSCs) impose several restrictions on the form and function of Merge, the only structure building operation in human language. To be more specific, Chomsky (2024) claims that Principle [S] requires Merge be subject to Minimal Search, a third factor principle that looks for a target of an operation with least effort, and thanks to this, Merge has exactly two subcases: External Merge (EM) and Internal Merge (IM). In the case of EM, Minimal Search picks out two syntactic objects P and Q in a workspace WS, generating the set {P, Q} in a new workspace WS' as in (2).[4]

[4] Workspace is defined by Chomsky (2021) and Chomsky et al. (2023) as a set of syntactic objects.

(2) External Merge (P, Q, WS) = WS', where

a. WS = [P, Q]

b. WS' = [{P, Q}]

In the case of IM, by contrast, Minimal Search looks for a syntactic object P in WS and then looks inside P, picking out another syntactic object Q within P. As a result, Merge yield the set {P, Q} in WS' as in (3).

(3) Internal Merge (P, Q, WS) = WS', where

a. WS = [{$_P$ Q, R}]

b. WS' = [{Q, {$_P$ Q, R}}]

Being a binary set-forming operation, Merge defines exactly three fundamental syntactic relations: sister-of, term-of, and c-command:

(4) a. X is a sister of Y iff X and Y are co-members of a set.

b. X is a term of Y iff either X is an element of Y or X is an element of a term of Y.

c. X c-commands Y iff X is a sister of Z and Y is a term of Z.

Sister-of in (4a) and term-of in (4b) are the simplest relations directly follows from set-theoretic notion of Merge: If we have the set γ = {α, β}, α and β are, by definition, co-members of γ. Also, α and β are, by definition, elements of γ. The term-of relation in (4b) is then recursively defined by the set-element relation: For example, if we have the set δ = {γ, {α, β}}, α is a term of δ because it is an element of the set {α, β}, which is an element of δ.[5] C-command in (4c) is the next simplest relation derived from combination of sister-of and term-of relations. For example, if we have the set {γ, {α, β}}, γ c-commands α because γ is the sister of {α, β} and α is a term of {α, β}. Thus, these three relations follow from Merge as a binary set-forming operation. As Chomsky (2024) notes, if language is optimally designed, it takes advantage of these and only these simplest relations that results from Merge.

Along with Principle [S], Principle [T] requires that Merge be associated with theta-structure interpreted at CI. In the case of EM, the output must be theta-related: the set {P, Q} created by EM is a pair of an assigner and an assignee of theta-roles.[6] In the case of IM, by contrast, the input must be

In this formulation, Merge is understood as mapping of WS onto a new workspace WS'.

[5] See also Chomsky et al. (2023: 19) for definition of term-of.

[6] Functional categories and adjuncts are put aside here (Chomsky (2024: 23, fn. 17)). For a solution to this problem, see Kitahara and Seely (2025).

theta-related: Only a member of a theta-structure is eligible to IM (Table 1).

Table 1

	External Merge	Internal Merge
Principle [S]: Minimal Search	P and Q are members of WS	P is a member of WS; Q is its term
Principle [T]: Theta Theory	P and Q are a theta-assigner and a theta-assignee	Q is a theta-assignee

This severely restrictive theory of Merge entails that once an item has undergone IM from a theta-position to a non-theta position, it never moved out of there via IM (that is, there is no successive-cyclic movement in this framework). To see this, consider the derivation of *What did you buy?* in (5).

(5) C_Q [you T [~~you~~ [what v* [V ~~what~~]]]]

First, *buy* and *what*, a pair of a theta-assigner and an assignee, externally merges. Second, the set is externally merged with v*. Third, *what* in the theta-position undergoes IM to Spec-v*, with the v*P phase closed.[7] Fourth, *you* is externally merged in the theta position, the sister of v*P. Fifth, T is introduced by EM and *you* is raised to Spec-T by IM. Finally, the interrogative complementizer C_Q is introduced by EM and the CP phase completes. Crucially, *what* raised to the v*P edge does not undergo further successive-cyclic IM to Spec-C as assumed in the previous minimalist frameworks like Chomsky (2021) and Chomsky et al. (2023). This is because IM is applicable only to a theta-marked item due to Principle [T], and *what* raised to the phase edge is no longer in a theta-position (For expository purpose, we put a framed box around an item out of a theta-position and inaccessible to Merge).[8]

An item escaped from a theta-domain, however, must still be available for interpretation related with illocutionary force, information structure, scope, etc. Chomsky (2024) proposes that at the CP phase-level, C accesses the item in the phase edge for instruction on how to interpret it at CI and how to pronounce it at externalization, as schematized in (6).

(6) C_Q [you T [~~you~~ [$\boxed{\text{what}}$ v* [V ~~what~~]]]]

[7] Chomsky (2024: 34: fn. 31) stipulates that the inner Spec-v* created by IM belongs to the v*P phase, whereas the outer Spec-v* created by EM of an external argument enters into the next phase.
[8] Notice that the square box is used only for ease of exposition: There is no operation like ‘boxing’ to mark an item that has undergone IM.

As a result, the *wh*-phrase serves as a matrix *wh*-operator and is pronounced at the left periphery of the clause.

Behind this analysis is the observation that language structures are relevant to two categories of thought, propositional structures related with theta-theoretic interpretation and clausal ones associated with illocutionary force, information structure, scope, etc., which are strictly separated from each other. This is called Duality of Semantics. The primary function of Merge is to produce these two structures: EM builds a propositional domain, whereas IM carries an argument from a propositional domain to a clausal one. In (6), for example, once *what* has escaped from the theta domain, it is rendered immune to theta-theoretic interpretation and unavailable to syntactic derivation — metaphorically speaking, it is in a 'box.' However, this argument becomes accessible to discourse-related interpretation at the clausal domain. Thus, once *what* has raised to Spec-v*, it is available for interpretation at the matrix CP phase as a reflex of association with the phase head C via c-command. As a result, *what* in Spec-v* is construed at CI as having scope over the matrix clause, with *what* externalized at the left periphery.

Chomsky (2024) extends his analysis to a subject *wh*-question in the following way. He assumes that a phase edge has a privileged status in that it is accessible to C regardless of whether it is the inner Spec-v* created by IM or the outer Spec-v* created by EM, the predicate internal subject position associated with theta-marking. Given this, the subject *wh*-question in (7a) is derived as in (7b).

(7) a. Who does it seem [*t* ate sandwich]? (Chomsky (2024: 34))

b. C_Q it seems [C [who T [~~who~~ [v* [V sandwich]]]]]

In (7b), *who* in Spec-T is not accessible to the matrix C_Q, whereas the one in Spec-v* is, because only the latter is in the phase edge. Then, C_Q accesses *who* in Spec-v*, externalizing it in the matrix Spec-C_Q.

Chomsky's analysis of subject *wh*-question, however, is not without a problem. As pointed out by Kitahara and Seely (2025), it undergenerates the sentence with passivization in (8a), whose structure is shown in (8b).[9]

(8) a. Who was arrested?

b. C_Q [who T [v [V ~~who~~]]]

[9] This was originally pointed out by Riny Huijbregts (Kitahara and Seely (2025: 13, fn. 28)).

Since a passive vP does not constitute a phase, *who* is raised from its theta-position to Spec-T in one-fell-swoop. So, neither *who* in Spec-T nor *who* in the sister of *arrest* is accessible to C_Q because neither counts as a phase edge. Thus, Chomsky's assumption that C_Q can only access an item in a phase edge fails to generate (8a). This problem is solved by Kitahara and Seely's (2024) analysis outlined in the next subsection.

2.2 Kitahara and Seely (2025)

Kitahara and Seely (2025) propose a revised framework that eliminates the notion of phase edge. They propose that (i) an argument undergoes EM and then IM to Spec-V/T, regardless of whether it is a *wh*-phrase or not, and (ii) such IM-ed element, escaped from a theta-position (i.e, being 'boxed'), is accessible to a higher phase head at later phase-level interpretation. Given this, *Who did Mary saw?* is derived as in (9).

(9) C_Q [$\boxed{\text{Mary}}$ T [~~Mary~~ [v* [$\boxed{\text{who}}$ [V ~~who~~]]]]]

In (9), both the internal argument (IA) *who* and the external argument (EA) *Mary* are obligatorily raised to Spec-V and Spec-I, respectively. Just after C_Q is introduced, the derivation reaches the CP phase level. Then, C_Q accesses *who* in Spec-V based on c-command relation between C_Q and *who*, externalizing it in Spec-C_Q and assigning it a matrix scope. Kitahara and Seely (2025) derives the subject *wh*-question *Who saw Mary?* as in (10).

(10) C_Q [$\boxed{\text{who}}$ T [~~who~~ v* [$\boxed{\text{Mary}}$ [V ~~Mary~~]]]]

Since the subject *wh*-phrase is raised from Spec-v* to Spec-T, it is rendered accessible to C_Q. When the derivation reaches the CP phase level just after C_Q is introduced, C_Q accesses *who* in Spec-T, externalizing *who* in Spec-C_Q and assigning it a matrix scope.

Kitahara and Seely (2025) argues that this analysis has advantages over Chomsky's (2024) in three respects. First, it provides a straightforward solution to Chomsky's (2024) problem in deriving the passive *wh*-question in (8a). Consider the derivation of (8a) illustrated (11).

(11) C_Q [$\boxed{\text{who}}$ T [v [V ~~who~~]]]

In (11), the IA *who* is directly raised from the sister of V to Spec-T. C_Q then accesses *who* in Spec-T, with *who* assigned a matrix scope. Thus, (8a) is successfully derived. Second, it eliminates the stipulation that of a v*P phase edge has a privileged status in that it is accessible to C. Third, it eliminates the stipulation that the inner Spec-v* belongs to the v*P phase, whereas the outer Spec-v* enters into the next phase (see footnote 6). In their framework, the notion of phase edge being eliminated, a v*P phase closes once v* is introduced. It follows that an external argument merged in the sister of v*P belongs to the next phase. Similarly, a CP phase closes just after C is introduced.

3. Island Conditions

Based on the MC framework outlined in section 2, we propose constraints on phase-level access from C, namely the Box-over-Box Condition and the Minimality Condition on Access. These two conditions, in effect, requires that C and an item accessible to C (i.e., a 'boxed' item) be as 'close' as possible. We will then show that these two conditions unify locality constraints on *wh*-movement such as the Subject Condition (section 3.1), the Adjunct Condition (section 3.2), the *Wh*-Island Condition (section 3.3) and the Complex NP Condition (section 3.4). Section 3.5 is an interim summary.

3.1 Constraints on Access

In this subsection, we purpose two conditions that constrain phase-level access. Before submitting our proposal, however, we must clarify what qualifies access from C in our framework.

Recall that in Chomsky (2024), accessibility to C is defined in terms of the notion phase-edge: C accesses only an item in the v*P phase edge. By contrast, Kitahara and Seely (2025) defines it in terms of phase-internal IM: C accesses an item raised from a theta-position (the sister of V or v*P) to a non-theta position (Spec-V/T). We assume that C accesses an element escaped from its base potion (i.e., the sister of a lexical head or the sister of a v*P phase). Our assumption is basically same with Kitahara and Seely's (2025) in that an IM-ed element is 'boxed' and accessible to C. We extend their analysis, claiming that not only a phrase escaped from a sister of V but also a phrase exited from a sister of any head is 'boxed' and hence accessible to C. To see how this works, consider (12), which have the structures in (13).

(12) a. What did John hear?
b. Who did John hear [stories about *t*]?
c. Who heard the story?

(13) a. C_Q [$\boxed{\text{John}}$ T [~~John~~ v* [$\boxed{\text{what}}$ [V ~~what~~]]]]

b. C_Q [$\boxed{\text{John}}$ T [~~John~~ v* [V [stories [$\boxed{\text{who}}$ [P ~~who~~]]]]]]

c. C_Q [$\boxed{\text{who}}$ T [~~who~~ v* [V Bill]]]]

In (13a), *what* is raised from Comp-V to Spec-V. Since it has escaped from the sister of V, it is accessible to C_Q. In (13b), *who* in Comp-P is raised to Spec-P. Having exited from the sister of P, *who* in Spec-P is qualified for access from C_Q. In (13c), *who* in the theta-position (i.e., the sister of v*P phase) is raised to Spec-T, and *who* in Spec-T is accessible to C_Q.

With these assumptions in place, we propose the condition in (14), referred to as the Box-over-Box (B-over-B) Condition.

(14) Box-over-Box Condition
If both XP and YP are qualified for access from C, and XP is a term of YP, then C does not access XP.

Informally speaking, the B-over-B Condition states that once search initiated by C has located a 'boxed' item, then it never 'look inside' the 'box.' In other words, the B-over-B Condition states that if there are two possible targets of access (i.e., 'boxed' items), only the 'closest' one is available for C. Here, closeness is defined in terms of term-of relation: If XP is a term of YP, then YP constitutes an opaque domain for access. Schematically, if we have the configuration in (15), where XP is a term of YP, C_Q never access XP for instruction (In (15), the square box and the mask stand for 'boxed' categories potentially accessible to C_Q).

(15) C_Q …… $\boxed{[_{YP} \ldots\ldots\ \text{XP}\ \ldots\ldots]}$

Second, we propose the condition in (16), referred to as the Minimality Condition on Access, where SD(C) stands for the search domain (i.e., the sister) of a phase head C.

(16) Minimality Condition on Access
If both C_1 and C_2 are associated with clausal interpretation, and the SD(C_2) is a term of SD(C_1), then C_1 does not access anything within SD(C_2).

Informally speaking, the Minimality Condition on Access bars search initiated by C from 'looking inside' the SD of another C. In (16), again, term-of relation is used to define the locality domain: the SD(C_2) serves as an opaque domain for access from C_1 when SD(C_2) is a term of SD(C_1), as schematically represented in (17).

(17) C_1 [$_{SD(C1)}$ …… C_2 [$_{SD(C2)}$…… $\boxed{XP}$ ……]]

In other words, The Minimality Condition on Access states that if a 'boxed' item is potentially accessible two CP phase heads, C_1 and C_2, it may provide 'instruction' only to the 'closest' one, where closeness is defined in terms of term-of relation between the two SDs. Notice also that (16) blocks phase-level access with the proviso that C_1 and C_2 should be phase heads that brings about semantic/pragmatic interpretation at the clausal domain. That is, when C_2 in (17) is not a head associated with illocutionary force, information structure, or scope, SD(C_2) counts as a transparent domain for access from a higher phase head.

The B-over-B Condition and the Minimality Condition on Access, conspiring together, require that C and a 'boxed' XP be as 'close' as possible: Thanks to the B-over-B Condition, C can only choose the 'closest' XP, whereas due to the Minimality Condition on Access, XP must provide an 'instruction' only to the 'closest' phase head C, thereby minimizing the search space for access. Notice that also that closeness here is defined in terms of term-of, which naturally arises from the Merge-based framework as discussed in section 2.1. In this sense, our proposal adheres the SMT, because it defines locality domains based only on the simplest relations that follow from Merge.

3.2 The Subject Condition

3.2.1 Extraction out of a Subject

In this subsection, we will show that the B-over-B Condition accounts for the Subject Condition in (18b) and (19b), where extraction out of the subject results in degradation in acceptability.

(18) a. Who did John hear [stories about *t*]?

b. * Who did [stories about *t*] terrify John? (Chomsky (1973: 106))

(19) a. Who did you visit [friends of *t*]?

b. ?*Who did [friends of *t*] hit Bill? (Lasnik and Saito (1993: 12))

(19b) has the structure in (20).

(20) C_Q [[friends [who [P ~~who~~]]] T [~~friends [who [P who]]~~ [v* [V Bill]]]]]

In (20), the external argument *friends of who* is raised from Spec-v* to Spec-T (Notice that before EM of the external argument into Spec-v*, *who* has already been raised to Spec-P and 'boxed'). When the derivation reaches the matrix CP phase-level just after C_Q is introduced, C_Q tries to find out an accessible *wh*-phrase within its SD. However, C_Q cannot locate *who* due to the B-over-B Condition because *who* is a term of the set *friends of who*, which is 'boxed' thanks to raising to Spec-T. Thus, we cannot derive (19b) by accessing *who*.

3.2.2 Extraction out of a Passive Subject and an Object

This analysis also accounts for asymmetry between extraction out of an object vs. a passive subject. As illustrated in (21) and (22), a passive subject does not allow subextraction, whereas an object does.[10]

(21) a. * Which cars were [the hoods of *t*] damaged by the explosion?
b. Which cars did the explosion damage [the hoods of *t*]? (Ross (1986: 148, fn. 30))

(22) a. ?*Who was [a picture of *t*] selected?
b. Who did you select [a picture of *t*]? (Lasnik (2001: 110-111))

The structure of (22a) is illustrated in (23).

(23) C_Q [[pictures [who [P ~~who~~]]] T [v [V ~~pictures [who [P who]]~~]]]]

The passive subject *pictures of who* is raised from Comp-V to Spec-V, thereby 'boxed.' Thus, accessing *who* to derive (22a) violates the B-over-B Condition.

In contrast, (22b) is derived as in (13) without violating the B-over-B Condition.

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[10] In the literature, it is sometimes observed that a passive subject allows subextraction:
(i) Of which cars were [the hoods *t*] damaged by the explosion? (Ross (1986: 148))
(ii) a. Of which car was [the (driver, picture) *t*] awarded a prize?
b. * Of which car did [the (driver, picture) *t*] cause a scandal? (Chomsky (2008: 125))
However, as Haegeman et al. (2014) observes, some independent factors such as pied-piping and D-linking lead to amelioration in acceptability of extraction out of a passive subject:
(iii) a. ??Of which famous celebrity were [compromising pictures *t*] published?
b. ?*Which famous celebrity were [compromising pictures of *t*] published?
c. * Who were [compromising pictures of *t*] published? (Haegeman et al. (2014: 125))
This observation leads us to speculate that (i) and (ii) is judged acceptable because of pied-piping and D-linking, though the question is left open how the paradigm in (iii) is accounted for.

(24) C_Q [you T [~~you~~ v* [V a picture [who [P ~~who~~]]]]]

A crucial assumption here is that object raising is optional (Lasnik (2001)). In (13), the object *a picture of who* stays in Comp-V and does not raise to Spec-V. Then, *who* raised to Spec-P is qualified for access from C_Q, because it is not a term of a constituent accessible to C_Q. Therefore, we can successfully derive (13) without violating the B-over-B Condition.

According to Lasnik (2001), optionality of object raising is empirically supported by the following verb-particle construction, where the object may be placed either before or after the particle.

(25) a. Mary called up [friends of John].
b. ? Mary called [friends of John] up. (Lasnik 2001: 111)

Lasnik shows that the object in the verb-particle-object (VPO) order like (26a) allows extraction out of the object, whereas the verb-object-particle (VOP) order like (26b) does not.

(26) a. Who did Mary call up [friends of *t*]?
b. ?*Who did Mary call [friends of *t*] up? (Lasnik 2001: 111)

(26) is accounted for if the object in the VPO order remains in Comp-V as in (27a), whereas the one in the VOP order undergoes raising to Spec-V as in (27b).[11]

(27) a. C_Q [you T [~~you~~ v^*_{call} [V_{up} a picture [who [P ~~who~~]]]]]

b. C_Q [you T [~~you~~ v^*_{call} [a picture [who [P ~~who~~]] [V_{up} ~~a picture [who [P who]]~~]]]]]

In (27a), C_Q accesses *who* without violating the B-over-B Condition. In (27b), since *a picture of who* undergoes object raising to Spec-V and it is 'boxed,' C_Q cannot access *who* within it due to the B-over-B Condition. Thus, we cannot derive (26b).

3.2.4 Extraction out of an ECM Subject

The proposed analysis can be extended to ban on extraction out of an ECM subject in (28) and (29).

[11] The assumption here is that the verb *call* undergoes head-raising from V to v*, whereas the particle *up* remains in V.

(28) * Who do you expect [stories about *t*] to terrify John? (Chomsky (1973: 106))

(29) a. Which artist do you admire [paintings by *t*]?

b. ?/*Which artist do you expect [paintings by *t*] to sell the best? (Polinsky (2013: 580))[12]

Assuming that an ECM subject is obligatorily raised to Spec-V (Postal (1974), Lasnik and Saito (1991), and Chomsky (2015)), (28) is derived as in (30).

(30) C_Q [you T [~~you~~ [v* [[stories [who [P ~~who~~]] V [to [~~stories [who [P who]]~~ [v* ...]]]]]]]

The ECM subject *stories about who* is raised from the embedded Spec-v*, a theta-position, to the matirx Spec-V, a non-theta one, thereby 'boxed.' Thus, the B-over-B Condition prohibits C_Q from accessing *who*. Thus, (28) is rule out as by the B-over-B Condition.

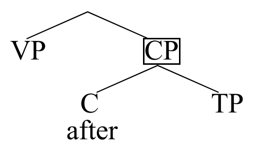
3.3 The Adjunct Condition

We next consider the Adjunct Condition in (31), where *wh*-movement out of an adjunct clause leads to degradation in acceptability.

(31) ?*Which book did John go to class [after he read *t*]? (Lasnik and Saito (1993: 12))

When an adjunct clause is merged with a modified constituent as in (32), the adjunct clause is not in the sister of a lexical head and hence it is not theta-marked. Since C accesses an element out of a sister of a head, the adjunct clause qualifies access from C. Thus, in (32), the adjunct CP counts as a 'boxed' category.

(32)



One may claim that this is inconsistent with the notion of Duality of Semantics, according to which a theta-structure is created by EM, whereas a discourse-related structure is by IM. In (32), the CP has not undergone IM but is available for the discourse/information-related interpretation at the CP phase. We suggest that this is due to Markovian property of derivation, a general property of derivation

[12] Polinsky (2013: 580) notes that (29a) is unproblematic, whereas (29b) is "marginal at best, and many native speakers reject this extraction altogether."

without looking back the derivational history (Chomsky (2021, 2024); Chomsky et al. (2023)). Since nothing keeps track of the history of derivation, we cannot know whether the adjunct CP in (32) is introduced by EM or IM when the derivation reaches the matrix CP phase level. Then, the adjunct CP, which is not in the sister of a head, is qualified for access from C irrespective of its previous derivational status.

Given this, (31) has the structure in (33) and violates the B-over-B Condition (EA and IA stand for the external and internal argument, respectively).

(33) C_Q [EA T [~~EA~~ [v* [V IA] [after [you T [~~you~~ [v* [which linguist [V ~~which linguist~~]]]]]]]]]

To derive (31), C_Q must access *which linguist* raised to Spec-V. However, it cannot be accessed by C_Q because it is a term of the 'boxed' adjunct clause. Thus, we cannot derive (31).[13]

3.4 The *Wh*-Island Condition

3.4.1 Extraction out of an Indirect Question

Let us consider the *Wh*-island Condition in (34), where extraction out of an indirect question results in degradation in acceptability.

(34) ??What did you wonder whether John bought? (Lasnik and Saito (1993: 72))

(34) has the structure in (35).

(35) C_Q [you T [~~you~~ v* [V [C_Q [John T [~~John~~ v* [what [V ~~what~~]]]]]]]]?

In (35), C_Q tries to access *what* in the embedded Spec-V. However, this violates the Minimality Condition on Access because the embedded complementizer C_Q is associated with interrogative force, and the SD of the embedded C_Q (i.e., the embedded TP) serves as an opaque domain for access from the matrix C_Q. More formally, since the embedded SD(C_Q) = {John {T, ...}} is a term of the SD of the

[13] Since the proposed analysis predicts that any adjunct clause merged higher than the complement of a verb constitutes an island, it might incorrectly rule out extraction out of 'high' adjuncts like (i), where rational clauses are merged higher than a v*P adverb and functional projections for progressive aspect and negation:

(i) a. What was she alternately going there every other Tuesday [to establish *t*]?
b. Who were you crossing (the street) so slowly [to (arrange to) meet *t*]?
c. What paper didn't you sleep [(in order) to finish *t*]? (Ernst (2022: 110))

We leave this problem for future research.

matrix SD(C_Q) = {you {T, ...}}, the former serves as the domain blocking the matrix C_Q from accessing anything within it.

Notice that the Minimality Condition on Access does not rule out long-distance *wh*-movement like (36), which has the structure in (37).

(36) What did you think that John bought?

(37) C_Q [you T [~~you~~ v* [V [C [John T [~~John~~ v* [$\boxed{\text{what}}$ [V ~~what~~]]]]]]]]?

In (37), the embedded clause is not headed by an interrogative complementizer but by a declarative one. Since a declarative C does not access a 'boxed' item within its SD to bring about force-related interpretation, SD(C) does not count as a domain blocking access from a higher phase head.

This analysis predicts that even the SD of a declarative C counts as a opaque domain when it is associated with information-structural interpretation. Consider (38).

(38) ??How do you think (that) this problem, John solved *t*? (Lasnik and Saito (1993: 96))

This sentence shows that *wh*-extraction is prohibited out of the embedded *that*-clause within which topialization takes place, which is known as the Topic Island. (38) has the structure in (39).

(39) C_Q [you T [~~you~~ v* [V [C [John T [~~John~~ v* [$\boxed{\text{this problem}}$ [$\boxed{\text{how}}$ [V ~~this problem~~]]]]]]]]]

In (39), the embedded declarative C accesses *this problem* to yield a topic-comment structure at the embedded CP phase. Then, accessing *how* within SD(C) = {John, {T, ...}} from the matrix C_Q violates the Minimality Condition on Access, because the embedded declarative C is associated with the topic and SD(C) is a term of SD(C_Q).

Similarly, this analysis accounts for the fact that that long-distance topicalization is sensitive to the *Wh*-island Condition, as shown in (40).

(40) ??That book$_i$, I wonder where$_j$ John put t_i t_j? (Lasnik and Saito (1993: 81))

(40) has the structure in (41).[14]

[14] I leave open the structure of a doble object construction in the MC framework and simply assume that *that book* and *where* is raised from its base positions to Spec-V by IM.

(41) C [I T [~~I~~ v* [V [C_Q [John T [~~John~~ v* [$\boxed{\text{that book}}$ [$\boxed{\text{where}}$ [V ~~that book~~ ~~where~~]]]]]]]]]

In (41), the embedded C_Q accesses *where*, and the matrix C accesses *that book* to yield a topic-comment structure. However, access from the matrix C to *where* violates the Minimally Condition on Access because SD(C_Q) is a term of SD(C).

3.4.2 The Proper Binding Condition Effect and the Freezing Effect

Let us consider the Proper Binding Condition effect in (42a) and the freezing effect in (42b).

(42) a. * [Which picture of t_i]$_j$ do you wonder who$_i$ John likes t_j?

b. ??Who$_i$ do you wonder [which picture of t_i]$_j$ John likes t_j? (Saito (1989: 187))

We first introduce Kitahara's (2025) analysis, which our analysis owes for insights. Then, I will propose a revised analysis based on the B-over-B condition and the Minimality Condition on Access.

In Kitahara's (2025) analysis, both (42a) and (42b) have the structure in (43), where *which picture of who* is raised to Spec-V, with *which picture of who* and *who* rendered accessible to C_Q.

(43) C_Q [you T [~~you~~ v* [V [C_Q [John T [~~John~~ v* [$\boxed{\text{which picture of who}}$ [V ~~which picture of who~~]...]

Kitahara (2025) argues that since C_Q accesses the 'closest' *wh*-phrase under Minimal Search, what is accessed by the embedded C_Q is not *who* but *which picture of who*. Notice that 'closeness' here is defined in terms of c-command: According to Kitahara (2025: 270), "[t]he closest *wh*-phrase to the C_Q is the one occupying Spec-V, because its head *which* c-commands *who*." As a result, deriving (42a) is impossible. By contrast, (42b) can be derived because (44) holds, by hypothesis:

(44) A *wh*-phrase, accessed in a previous phase cycle, does not interfere with subsequent search from a higher C_Q, because it has no relevant instructions to provide.

(adapted from Kitahara (2025: 271))

With this, the matrix C_Q successfully accesses *who* without intervened by *which* because the latter has already been accessed at the embedded CP cycle and is no longer available for the phase-level access at the matrix CP cycle. Thus, we can generate (42b), with *who* externalized at the matrix Spec-C_Q.

Although Kitahara's (2025) analysis elegantly account for why (42a) and (42b) differ in acceptability, there remains two questions unsolved. First, it leaves open whether there is independent empirical evidence for (44). Second, it remains to be seen why (42b) is not completely acceptable but results in weak, subjacency-level violation.

As for the first question, we suggest that (44) is supported by the Criterial Freezing effect (Rizzi 2006, 2010, 2015; Epstein et al. 2015) like (45), where once a *wh*-phrase has entered into agreement with an embedded C_Q, it does not undergo further *wh*-movement to the matrix CP.

(45) a. You wonder which dog John likes *t*.

b. * Which dog do you wonder John likes *t*? (Epstein et al. (2015: 230))

In the MC framework, (45a) and (45b) has the structure in (46a) and (46b), respectively.

(46) a. C [you T [~~you~~ v* [V [C_Q [John T [~~John~~ v* [which dog [V ~~which dog~~]]]]]]]]

b. C_Q [you T [~~you~~ v* [V [C_Q [John T [~~John~~ v* [which dog [V ~~which dog~~]]]]]]]]

In (46a), the embedded C_Q accesses *which dog*, assigning it the embedded scope. In (46b), *which dog* is accessed by the embedded C_Q to yield an embedded scope, and then accessed again by the matrix C_Q to assign it the matrix scope. If the double-access to *which dog* were allowed, we would overgenerate (45b). However, this is impossible because if (44) holds: Once *which dog* has been accessed at the embedded cycle, it is no longer available for interpretation at the matrix CP cycle.

As for the second question, we suggest that the freezing case in (42b) violates the Minimality Condition on Access. Consider the structure of (42b) illustrated in (47), where the 'short' raising of *who* to Spec-P is abstracted away for simplicity of illustration.

*Minimality

(47) C_Q [you T [~~you~~ v* [V [C_Q [John T [~~John~~ v* [which picture of who [V ~~which picture of who~~]...]

In (47), the matrix C_Q tries to access *who* but the embedded SD(C_Q) is a term of the matrix one. Therefore, (42b) results in subjacency-level degradation owing to the Minimality Condition on Access as the *wh*-island case in (36) does. Notice also that accessing from the matrix C_Q to *who* does not violate the B-over-B Condition, although *who* is a term of *which picture of who*. This is because *which picture of who* has already been accessed in the embedded CP cycle and is no longer available for the phase-level interpretation at the matrix CP cycle. As a result, *which picture of who* does not block the matrix C_Q from accessing *who*.

In our framework, the Proper Binding Condition case in (42a) is ruled out because it violates both the B-over-B Condition and the Minimality Condition on Access. Consider the structure of (42a) illustrated in (48), where raising of *who* to Spec-P is again abstracted away for simplicity of illustration.

* B-over-B

(48) C_Q [you T [~~you~~ v* [V [C_Q [John T [~~John~~ v* [which picture of who [V ~~which picture of who~~]...]

*Minimality

In (48), the matrix C_Q and the embedded one tries to access *which picture of who* and *who*, respectively. However, access from the embedded C_Q to *who* violates the B-over-B Condition, because it is a term of *which picture of who*. (48) also violates the Minimality Condition on Access: Since the embedded SD(C_Q) is a term of the matrix one, the former constitutes an opaque domain for access from the matrix C_Q. Thus, (42a) violates both the B-over-B Condition and the Minimality Condition on Access, thereby resulting in severe degradation in acceptability.

3.4.3 Freezing Effect in Topicalization

The proposed analysis also makes it possible to account for the freezing effect in topicalization shown in (49), where topicalization out of a topicalized phrase leads to subjacency-level degradation in acceptability.

(49) a. I think that you should read [articles about vowel harmony] carefully.
b. I think that [articles about vowel harmony], you should read *t* carefully.
c. ?? Vowel harmony, I think that [articles about *t*] you should read *t* carefully.

(Lasnik and Saito (1993: 101))

The structure of (49c) is shown in (50), with raising to Spec-P abstracted away.

*Minimality

(50) C [I T [~~I~~ v* [V [C [you T [~~John~~ v* [articles about vowel harmony [V ~~articles about vowel harmony~~]]]]]]]]

To derive (49c), the embedded C accesses *articles about vowel harmony* at the embedded CP cycle.

Then, the matrix C may access *vowel harmony* without violating the B-over-B Condition at the matrix CP cycle since *articles about vowel harmony* has already been accessed in the embedded phase cycle. However, *vowel harmony* is within the embedded SD(C), which is the term of the matrix one, and the embedded C is associated with discourse-structural interpretation. Therefore, access from the matrix C to *vowel harmony* violates the Minimality Condition on Access. Thus, (49c) results in subjacency-level degradation in acceptability.

3.5 The Complex NP Condition

Let us finally consider the Complex NP Condition like (51), where *wh*-extraction out of a relative clause leads to severe degradation in acceptability.

(51) * Where$_j$ did you see the book [which$_i$ John put t_i t_j] (Lasnik and Saito (1993: 71))

The structure of (51) is shown in (52).

(52) C_Q [you T [~~you~~ v* [V [the book [C_R [John T [John v* [which [where [V ~~which~~ ~~where~~]]]]]]]]]]

*B-over-B, *Minimality

In (52), the relative pronoun *which* is raised from the object position to Spec-V and is accessed by the C head of the relative clause (C_R) to bring about an operator-variable interpretation. Additionally, the matrix C_Q tries to accesse *where* in the relative clause. However, this access violates the B-over-B Condition because *where* is inside the relative clause, which is an adjunct modifying the noun *book*. Since C is accessible to any phrase which is not in a complement of a head irrespective of whether it is introduced by EM or IM, the relative clause serves as a category blocking access to anything within it. Additionally, access from C_Q to *where* violates the Minimality Condition on Access because the SD(C_R) is a term of SD(C_Q), and C_R is a phase head associated with the operator-variable interpretation established by access from C_R to *where*. Thus, (51) violates both the B-over-B Condition and the Minimality Condition on Access, thereby resulting in severe degradation in acceptability.

3.6 Interim Summary

Summarizing so far, we demonstrated that some island conditions fall within violation in the B-over-B Condition violation and/or the Minimality Condition on Access. Table 2 shows how these two conditions are violated by the examples discussed in this section.

Table 2

	Examples	B-over-B	Minimality
Extraction out of a subject	(18b), (19b)	*	OK
Extraction out of a passive subject	(21a), (22a)	*	OK
Extraction out of an object	(18a), (21b)	OK	OK
Extraction out of a shifted object	(26b)	*	OK
Extraction out of an ECM subject	(28), (29b)	*	OK
The Adjunct Condition effect	(31)	*	OK
Wh-Island effect	(34), (40)	OK	*
Long-distance *wh*-movement	(36)	OK	OK
The Topic Island effect	(38)	OK	*
The Proper Binding Condition effect	(42a)	*	*
The freezing effect	(42b)	OK	*
Extraction out of a topic	(49c)	OK	OK
The Complex NP Condition effect	(51)	*	*

4. Superiority Effects

In the previous section, we argued that both the B-over-B Condition and the Minimality Condition on Access define locality domains in terms of term-of relation, which naturally arises from the Merge-based framework. Notice that in our framework, c-command relation is not used in defining these locality conditions. This brings about a further question of how to account for intervention phenomena in A'-movement such as superiority effects in English multiple *wh*-questions, which have been standardly analyzed in terms of 'closeness' defined by c-command. In this section, we attempt to defend the view that c-command relation is not at work in defining locality domains of phase-level access. In section 4.1, we will see that elimination of the intervention condition defined by c-command bring about a welcome consequence in the MC framework to the effect that it removes a potential problem of undergeneration in *wh*-movement and topicalization. In section 4.2, we examine a tentative analysis of English multiple *wh*-questions in the MC framework which crucially relies on the notion of intervention, showing that this proposal faces an empirical problem. In section 4.3, we propose an analysis free from the notion of intervention. More specifically, we argue that multiple *wh*-questions in English is not derived by phase-level access from C_Q but by access from a *wh*-phrase already accessed by C_Q, demonstrating that this analysis accounts for superiority effects and their exceptions.

4.1 A Puzzle about the MC Framework

In this subsection, we will argue that elimination of intervention defined by c-command brings about a desirable consequence of solving a potential problem of undergereration in the MC framework.

To see this, consider the derivation of *What did you buy?*, which is represented as in (6), repeated here as (53), under Chomsky's (2024) framework.

(53) C_Q [you T [~~you~~ [what v* [V ~~what~~]]]]

Recall that in Chomsky's (2024) framework, C accesses anything in v*P edge, regardless of whether it is in the outer or inner Spec-v*. In (53), *you* in the outer Spec-v* asymmetrically c-commands *what* in the inner Spec-v*. Thus, if C_Q picks out an accessible item 'closest' to it, and closeness is defined in terms of c-command, *you* in the outer Spec-v* will block access from C_Q to *what* in the inner Spec-v*.[15]

The potential problem of undergeneration is not avoided in Kitahara and Seely's (2025) framework either, in which *What did you buy?* has the structure in (54).

(54) C_Q [you T [~~you~~ [v* [what [V ~~what~~]]]]]

According to Kitahara and Seely (2025: 15), "C_Q simply searches into its c-command domain and takes the first element." If the search procedure terminates once locates the 'boxed' *you* in Spec-T and it never search for anything within the c-command domain of *you*, C_Q fails to pick out *what*, thereby undergenerating *What did you buy?*.

One might claim that this problem does not arise if we assume *who* involves a dialectic A'-feature like [wh], and search from C_Q does not terminate unless it locates an element involving [wh]. However, this solution arouses a further problem in topicalization. Consider the structure of *This book, you should read.,* which has the structures in (55a) under Chomsky's (2024) framework and is represented as in (55b) under Kitahara and Seely's (2025).

(55) a. C [you T [~~you~~ [this book v* [V ~~this book~~]]]]

b. C [you T [~~you~~ v* [this book [V ~~this book~~]]]]

[15] This problem might not arise if we simply stipulate that the inner and outer Spec-v* is 'equidistant' from C, thought it must be clarified what is the proper definition of equidistance and whether this notion is derived from maximally simple structure building operations and optimal computation.

To derive the sentence with topicalization, C must access *this book* across the intervening element *you*, which asymmetrically c-command *this book*. The problem of intervention might be solved if we assume a topic feature [top] in *this book*. However, as Chomsky et al. (2019: 250) put, "informational notions such as "topic" or "focus," like grammatical functions or thematic roles, are properties of configurations and their syntactic/discursive context, not of individual syntactic objects" and "they should neither be represented in the lexicon, nor in the narrow syntactic derivation." As far as this is correct, we cannot postulate [top] in (55) to identify what C picks out. Instead, the topic-comment structure like [$_{\text{topic}}$ this book] [$_{\text{comment}}$ you should read x] is 'read-off' at CI from the configurations in (55), where *this book* has escaped from the theta position and provides instruction to at the CP phase level through c-command relation between C and *this book*. Thus, as far as we adopt the configurational approach to information structures, we cannot have recourse to the [top] feature to derive topicalization.

This puzzle is solved in our framework because c-commanding item does not serve as an intervener. For example, *What did you buy?* and *This book, you should read.*, which have the structures in (56), can successfully derived because in our system *you* and *we* in Spec-T do not prohibit C_Q from accessing *who* and *this book* in Spec-V although the former c-commands the latter.

(56) a. C_Q [$\boxed{\text{you}}$ T [~~you~~ [v* [$\boxed{\text{who}}$ [V ~~who~~]]]]]

b. C [$\boxed{\text{we}}$ T [~~we~~ v* [$\boxed{\text{this book}}$ [V ~~this book~~]]]]

Thus, the paradox discussed above does not arise in our system, and we can successfully derive (56) without assuming discourse-related/A-bar features like [wh] and [top].

If c-command plays no role in defining locality conditions, however, a question arises how the superiority effect in multiple *wh*-questions in (57a), which has the structure in (57b), is accounted for.

(57) a. * What did who see *t*?

b. C_Q [$\boxed{\text{who}}$ T [~~who~~ [v* [$\boxed{\text{what}}$ [V ~~what~~]]]]]

At first glance, the proposed system overgenerates (57a) because nothing in our system prevents C_Q from accessing *what* in Spec-V across *who* in Spec-T, which asymmetrically c-commands *what*. Thus, unless the structure in (57b) is ruled out on an independent ground, our system incorrectly rules in (57a). The next two subsections address this problem.

4.2 Superiority Effects in the MC Framework

In this subsection, we discuss the superiority effect in multiple *wh*-questions like (58) and (59), where the *wh*-in-situ intervenes overt *wh*-movement.

(58) a. Who saw what?

b. * What did who see *t*? (Lasnik and Saito (1993: 16))

(59) a. Who did you tell *t* to read what?

b. ?*What did you tell who to read *t*? (Lasnik and Saito (1993: 119))

(58) shows that the object *wh*-phrase cannot undergo overt *wh*-movement across the *wh*-phrase in the subject. Similarly, (59), which is sometimes referred to as the 'pure' superiority paradigm, shows that the *wh*-phrase in the embedded clause cannot be moved across the matrix *wh*-in-situ.

In the generative literature, the superiority effect in (58) and (59) has sometimes been accounted for in terms of the notion of closeness, which goes back to Chomsky's (1973) Superiority Condition.[16] In minimalism, assuming Chomsky's (1995) Minimal Link Condition, Kitahara (1997) proposes that C_Q attracts the 'closest' *wh*-phrase, namely *who*: In (58) and (59), *who* is closer to C_Q than *what* because the former c-commands the latter. Thus, under the Minimal Link Condition, only (58a) and (59a) are derivable. Along this line, one might attempt to analyze the superiority effect in the MC framework, proposing that access from C_Q is blocked by an intervening *wh*-phrase asymmetrically c-commanding a 'boxed' *wh*-phrase. For example, one might argue that (58a) is grammatical because there is no intervening *wh*-phrase between C_Q and *who* as schematized in (60a), whereas (58b) is ungrammatical because *who* blocks access from C_Q to *what* as illustrated in (60a), where *who* in Spec-T asymmetrically c-commands *what* in Spec-V.

(60) a. C_Q [$\boxed{\text{who}}$ T [~~who~~ [v* [$\boxed{\text{what}}$ [V ~~what~~]]]]]

b. C_Q [$\boxed{\text{who}}$ T [~~who~~ [v* [$\boxed{\text{what}}$ [V ~~what~~]]]]]

This approach, however, faces an empirical challenge: As observed by Lasnik and Saito (1993), acceptability of (58b) and (59b) is dramatically improved when they are embedded in *wh*-questions, as shown in (61).

(61) a. Who wonders [what who bought *t*]? (Lasnik and Saito (1993: 118-119))

b. Who wonders [what you told who to read *t*]? (Lasnik and Saito (1993: 121))

[16] See Dayal (2017) for previous approaches to multiple *wh*-questions.

This fact suggests that in (61), the embedded C_Q may access *what* in the embedded Spec-V to externalize it in the embedded left periphery and assign it an embedded scope across *who*, which asymmetrically c-commands *what*. That is, if we assume that access from C_Q is blocked by an intervening *wh*-phrase, we incorrectly rule out (61).

4.3 A Proposed Analysis

In this section, we propose an alternative analysis of superiority effects in English multiple *wh*-questions and lack thereof with no recourse to intervention in terms of c-command. More specifically, we propose (62).

(62) In English, a *wh*-phrase associated with C_Q optionally searches another *wh*-phrase within its SD to yield a double question reading.

The gist of this analysis is that a *wh*-in-situ is not accessed by C_Q but by a *wh*-phrase accessed by C_Q, which is externalized at the left periphery via access but remains in the 'boxed' position in syntax. To see how this works, consider the structure of (58a) illustrated in (63).

(63) C_Q [$\boxed{\text{who}}$ T [~~who~~ v* [V what]]]

In (63), C_Q accesses *who* in Spec-T, externalizing it in Spec-C_Q. Then, *who* in Spec-T, being accessed by C_Q, searches another *wh*-phrase within its SD, locating *what* in the object position.[17] As a result, (58a) brings about a double question reading. Let us next consider the structure of (58b) illustrated in (64).

(64) C_Q [$\boxed{\text{who}}$ T [~~who~~ v* [$\boxed{\text{what}}$ [V ~~what~~]]]]

In (64), C_Q accesses *what* in Spec-V, externalizing it at the left periphery. Notice that in our framework, nothing bars C_Q from accessing *what* in Spec-V across *who* in Spec-T. Then, *what* in Spec-V, being accessed by C_Q, tries to seek another *wh*-phrase within its SD. However, *what* cannot locate a *wh*-phrase because *who* in Spec-T is outside the SD of *what*. As a result, *who* in Spec-T left uninterpreted at CI, with (64) violating Full Interpretation. Thus, the subject-object asymmetry in (58a) is reduced

[17] We leave open whether a *wh*-in-situ is raised to Spec-V and 'boxed,'or it remains in the position where it is base-generated.

to the sizes of the SDs of the accessed *wh*-phrases.

Notice that the assumption in (62) that C_Q does not directly access a *wh*-in-situ is independently needed to account for the fact that in English, a *wh*-in-situ does not serve as an interrogative operator by itself; it must be licensed by another *wh*-phrase externalized at Spec-C_Q.[18] Consider (65), where the object *wh*-in-situ does not have an embedded scope unless another *wh*-phrase does.[19]

(65) a. * I wonder [John saw who].
 b. I wonder [who saw what]. (Lasnik and Saito (1993: 4-6))

If C_Q could directly access the *wh*-in-situ as in (66) to assign it an embedded scope at CI without externalizing it at the embedded Spec-C_Q, then we would incorrectly rule in (65a).

(66) C_Q [John T [~~John~~ v* [$\boxed{\text{who}}$ [V ~~who~~]]]]

In our framework, this problem does not arise because C_Q never directly access a *wh*-in-situ. Instead, an *wh*-in-situ has a scope as far as it is successfully licensed by another 'boxed' *wh*-phrase that has already been accessed by C_Q.

The proposed analysis also correctly predicts the contrast in (59a) and (59b), which have structures in (67a) and (67b), respectively.

(67) a. C_Q [you T [~~you~~ v* [$\boxed{\text{who}}$ [V ~~who~~ [C [PRO T [~~PRO~~ v* [V what]]]]]]]]
 b. C_Q [you T [~~you~~ v* [V who [C [PRO T [~~PRO~~ v* [$\boxed{\text{what}}$ [V ~~what~~]]]]]]]] X

In (67a), C_Q accesses *who* in the matrix Spec-V, followed by access from *who* to *what* in the embedded clause. As a result, (59) yields a double question reading. In (67b), C_Q accesses *what* in the embedded Spec-V. Although nothing bars C_Q from accessing *what* across *who*, *what* cannot locate *who* in the matrix clause since *who* is out of SD(*what*). As a result, *who* is left uninterpreted at CI.

Our analysis also accounts for the fact that an indirect question shows the superiority effect as in (68a), whereas its acceptability is dramatically improved when the matrix clause have an additional *wh*-phrase as in (61a), repeated here as (68b), where the embedded *who* does not have a scope at the

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[18] In English, *wh*-in-situ is not freely available but allowed only in multiple *wh*-questions and echo questions. We put aside echo-questions here.
[19] In the GB framework, this is attributed to the S-structure filter according to which a *wh*-phrase must occupy a [+WH] COMP (Lasnik and Saito (1984, 1994)).

embedded clause but at the matrix one.

(68) a. * Bill wonders [what who bought *t*]?

b. Who wonders [what who bought *t*]? (Lasnik and Saito (1993: 118-119))

The structure of (68a) is shown in (69).

(69) C [Bill T [~~Bill~~ v* [V [C_Q [$\boxed{\text{who}}$ T [~~who~~ [v* [$\boxed{\text{what}}$ [V ~~what~~]]]]]]]]]

In (69), the embedded C_Q accesses *what* in the embedded Spec-V. Then, *what* cannot locate *who* in the embedded Spec-T, thereby *who* left uninterpreted at CI. By contrast, (68a), whose structure is shown in (70), does not cause any problem at CI.

(70) C_Q [$\boxed{\text{who}}$ T [~~who~~ v* [V [C_Q [$\boxed{\text{who}}$ T [~~who~~ [v* [$\boxed{\text{what}}$ [V ~~what~~]]]]]]]]]

In (70), the matrix C_Q and the embedded one accesses *who* in the matrix Spec-T and *what* in the embedded Spec-V, respectively. Since the embedded *who* in the embedded Spec-T is outside the SD of *what*, it cannot bring about a double-question reading at the embedded clause through access from *what*. However, since the embedded *who* is within the SD of *who* in the matrix Spec-T, it is accessed by the matrix *who* to yield a double-question reading. Thus, (70) is successfully interpreted at CI, with *who* in the embedded Spec-T having a scope at the matrix clause thanks to the association with the matrix *who*.

This analysis straightforwardly explains improvement in acceptability in the ‘pure’ superiority paradigm in (61), repeated here as (71), where the embedded *who* have only the matrix scope.

(71) Who wonders [what you told who to read *t*]?

(71) has the structure in (72).

(72) C_Q [$\boxed{\text{who}}$ T [~~who~~ v* [V [C_Q [you T [~~you~~ v* [V who [C [PRO T [~~PRO~~ v* [$\boxed{\text{what}}$ [V ~~what~~]]]]]]]]]]]]

In (72), the embedded C_Q accesses *what* in Spec-V, externalizing it at the embedded Spec-C_Q. Although *who* in the embedded Spec-T is outside the SD of *what*, it is within that of *who* in the matrix Spec-T, which is accessed by the matrix C_Q. Thus, (72) is successfully interpreted at CI, with the embedded *who* having a matrix scope.

In this section, we argued that multiple *wh*-questions in English is derived by access from a *wh*-phrase already accessed by C_Q. This analysis, as far as it is on the right track, reinforces our claim that access from C is not constrained by intervention defined by c-command.

5. Conclusion

To conclude, based on the MC framework proposed by Chomsky (2024) and developed by Kitahara and Seely (2025), we argued that phase-level access from C is subject to two locality conditions, the B-over-B Condition and the Minimality Condition on Access. The former states that a 'boxed' item must be as 'close' to C as possible, whereas the latter requires C be 'close' to a 'boxed' item as far as possible, where closeness is defined in terms of term-of relation that 'comes free' from the Merge-based system. We also demonstrated that these two conditions unify constraints on *wh*-movement in English, such as the Subject Condition, the Adjunct Condition, the *Wh*-Island Condition, and the Complex NP Condition. Since these two conditions does not rely on c-command relation in defining locality domains, the proposed system, at first glance, seems to fail to account for superiority effects in English multiple *wh*-questions. To address this issue, we proposed that multiple *wh*-questions in English are not derived by phase-level access from C_Q but by access from a *wh*-phrase already accessed by C_Q, demonstrating that superiority effects and lack thereof are accounted for with no recourse to the notion of intervention defined in terms of c-command.

Our proposal, if on the right track, has three theoretical implications. First, the proposed analysis adheres as closely as possible to the SMT since it reduces locality constraints on *wh*-movement to general principles to minimizes the search space for access while having recourse only to the syntactic relations that follows from Merge. Second, since our proposal does not rely on A'-features like [wh] and [top] to derive *wh*-movement and topicalization, it suggests that these features are eliminable; at least they play no role in triggering displacement. Instead, the proposed analysis defended the view that these discourse-related properties are 'read-off' from the configurations generated by IM and EM. Finally, since the proposed analysis does not use the notion of c-command in defining locality domains of *wh*-movement, it might open a way to eliminate the notion of intervention in A'-dependencies, which has sometimes been assumed in generative literature (e.g., Relativized Minimality, the Minimal Link Condition, Probe-Goal Search, among others).

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